ex:- $A_1 = \{-1, 0, 1\}$ $A_2 = \{-2, 0, 2\}$ $A_3 = \{-3, 0, 3\}$

$\dots$ $\boxed{A_i = \{-i, 0, i\}}$

$\bigcup_{i=1}^{4} A_i = \{-4, -3, \dots, 0, \dots 3, 4\}$ $\bigcap_{i=1}^{4} A_i = \{0\}$

$\bigcup_{i=1}^{\infty} A_i = ? \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots\} = \mathbb{Z}$

$A_1 \cup A_2 \cup A_3 \cup A_4 \dots$

$\{-1, 0, 1\} \cup \{-2, 0, 2\} \cup \{-3, 0, 3\} \cup \{-4, 0, 4\} \dots$

$\bigcap_{i=1}^{\infty} A_i = \{0\}$ $\bigcup_{i=1}^{3} A_i = \bigcup_{i \in \{1,2,3\}} A_i = A_1 \cup A_2 \cup A_3$

Ex:- $A_\alpha = [\alpha, 2] \times [2, \alpha]$

$\bigcup_{\alpha \in [0,2]} A_\alpha = ?$ $\bigcap_{\alpha \in [0,2]} A_\alpha = ?$